Unit 3

Campbell Biology, AP Edition, 12e

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Genetics

Dr. Francisco Mojica

Dr. Francisco Mojica is a Professor of Microbiology at the Universidad de Alicante in Spain, where he studies CRISPR sequences. His parents fostered in him a love of nature, and he realized in high school that he was most interested in biology. In college, when he first looked through a microscope and saw small single-celled “living beings” moving around, he was captivated, and he has spent his career focusing on prokaryotes. While in graduate school in the 1990s, he found the prokaryotic DNA repeats that he later

named CRISPR repeats, and he spent the next decade establishing their role as a prokaryotic immune system. Dr. Mojica's lab currently studies the mysteries that still surround CRISPR, trying to understand every single aspect of its mechanism of action and regulation.

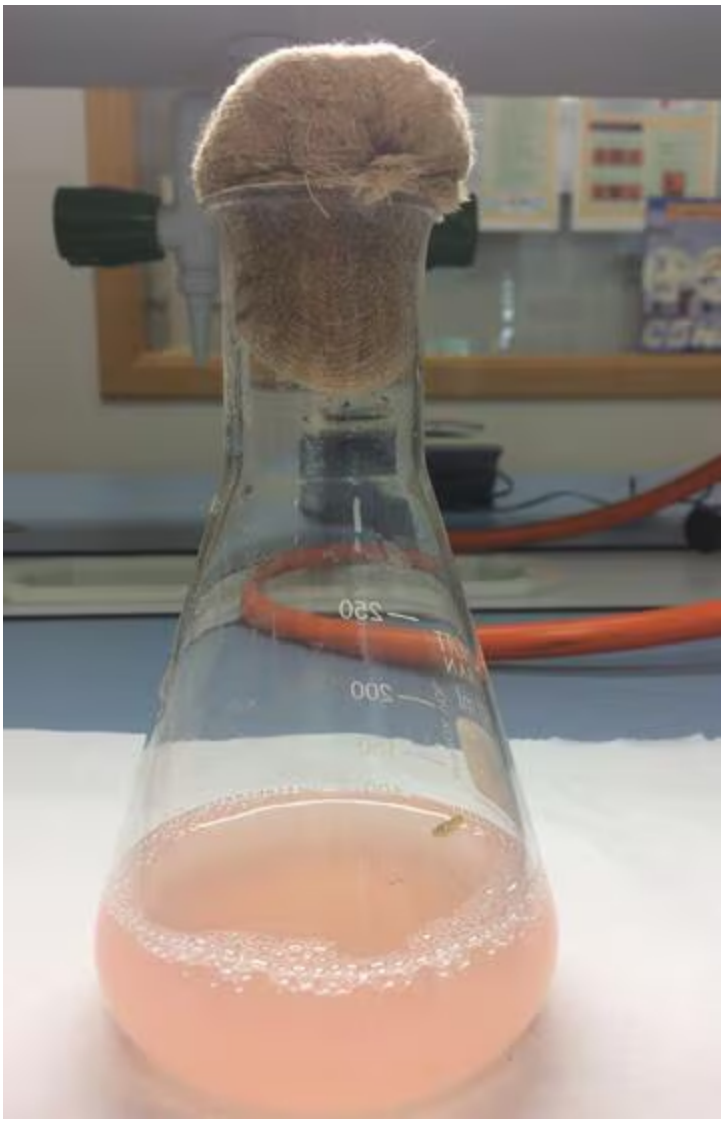
An Interview with: Francisco Mojica

When did you first decide to be a biologist?

In high school, my parents asked me, "What are you thinking you'll do with your life—become a doctor or a lawyer?" I said, "Not at all. I really like biology." In spite of his concern about my future career, my father told me I must study what I loved, so that I could be happy in my life. I really appreciated his words! So I went to university and studied biology. When you like biology, initially you get attracted because of the things you see—animals and plants. But when you see microscopic living beings for the first time, and they're moving around, that's really amazing! So, I decided to focus on microbiology.

What did you study as a graduate student?

I studied prokaryotes that live in salt ponds. These weird organisms can tolerate very high salt concentrations. I was interested in doing molecular biology, which had not been done in that university before—I knew that would be a challenge, and I love challenges. There weren't many genome sequences available at that time, in the early 1990s, but we had cloned a few sequences, and we had indirect evidence they might be related to adaptation to salinity. One of the DNA sequences had these regularly spaced repeats that were partially palindromic (reading the same forward and backward), so they could form secondary structures. We thought maybe these repeats were affected by salinity and would in turn affect gene expression, but we were absolutely wrong! At the end of my Ph.D. thesis, all I could say was, "We have found regularly interspaced repeats that are expressed by the cell, and they could be very important."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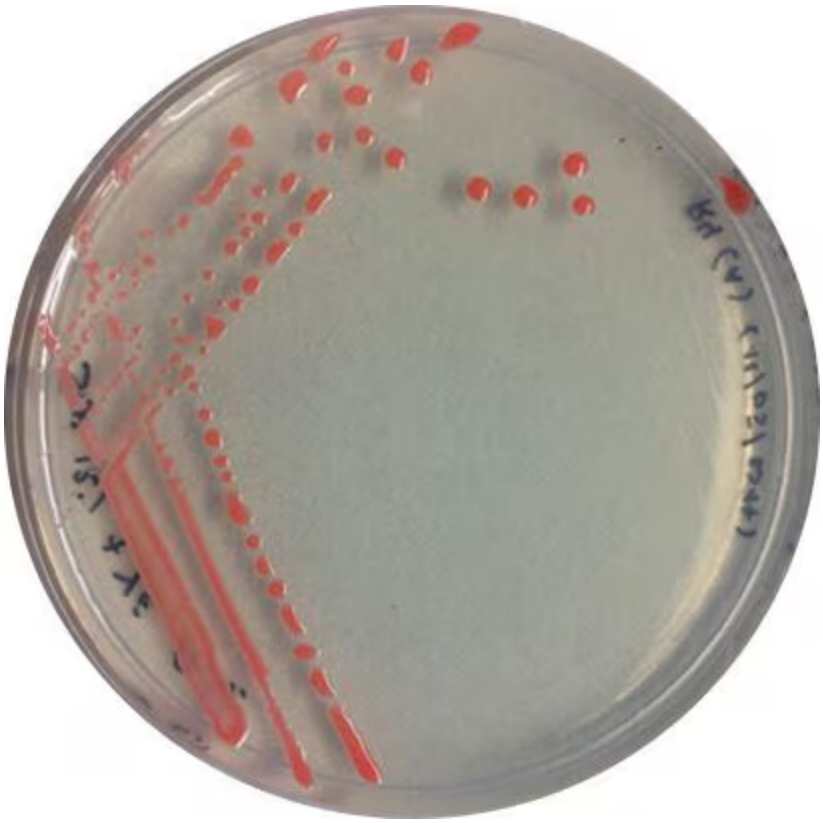

Dr. Mojica cultured and analyzed prokaryotes from this salt lake in Spain, leading to his discovery of CRISPR.

What made you continue to study these repeats?

When you find something in your thesis, you believe that it belongs to you—you have to try to answer that question. No one could figure out the role that these repeats played. However, others had found similar repeats in bacteria like *E. coli* and another very distantly-related bacterial species. That was an important finding that pushed me to keep working on the repeats, because it meant the repeats must be doing something important for them to stay in the genomes of various prokaryotes over such long evolutionary time. Evolution is so fundamental to biology, to understanding everything.

How did you figure out what the repeats were doing?

I started wondering about the sequences *between* the repeats, the so-called “spacers.” Where did they come from? I compared the spacer sequences to genome database sequences. By 2000 or 2001, many more genome sequences were available. Finally, in 2003, I found a spacer in *E. coli* whose sequence was identical to the sequence of a virus that infects *E. coli*. Even more interesting, the strain of *E. coli* with that spacer could not be infected by that virus. I checked more and more spacer sequences, and when a spacer sequence matched a sequence in the database, the match was always to the sequence of a virus that could no longer infect that species. So I realized these bacteria keep a “memory” of the viruses that infect them, and that the

repeat regions were part of an immune system—an unexpected and very surprising finding. I proposed naming the repeats Clustered Regularly Interspaced Short Palindromic Repeats, or CRISPR, and to my surprise, the name took hold.

> ***"Studying biology is basically just saying 'Wow!' all the time."***

What is your advice to a student considering a career in biology?

Studying biology is basically just saying "Wow!" all the time. In science, the truth changes very fast. The most important thing is to be open. Be ready to hear something that destroys what you believe—because biology is a fast-moving field, a field that evolves a lot. Knowledge will change.